METRIC

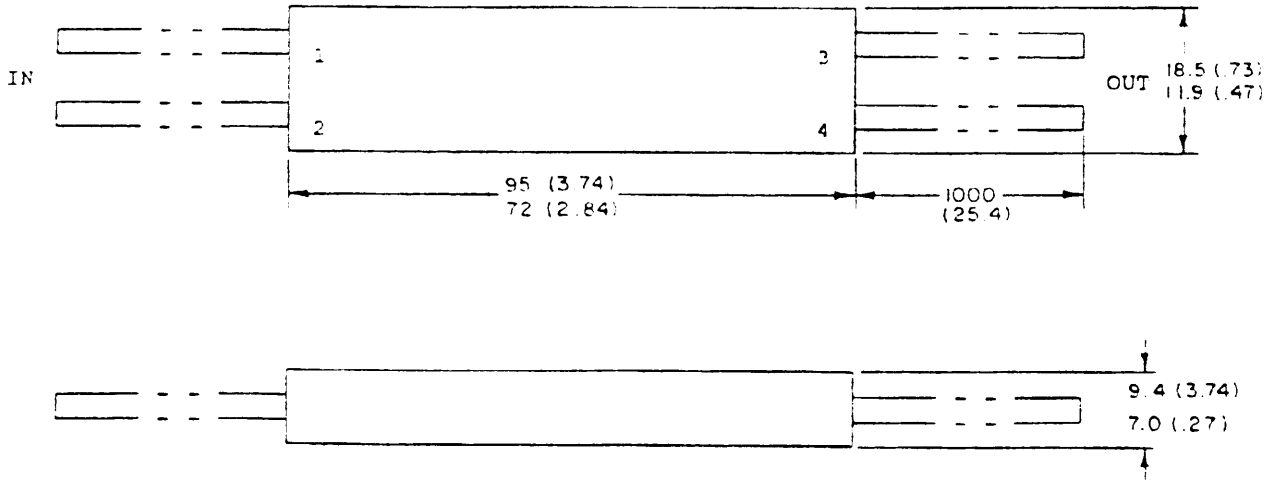
MIL-C-24621/5A
6 June 1991
SUPERSEDING
MIL-C-24621/5 (NAVY)
16 MAY 1989

MILITARY SPECIFICATION SHEET

COUPLERS, FIBER OPTIC CABLE, CABLE SPLITTER, PASSIVE GLASS-PIGTAILED OUTPUT, TYPE II, 1280 TO 1360 NANOMETER WAVELENGTH RANGE, 2 X 2 SYMMETRICAL TRANSMISSION STAR, 62.5/125 MICROMETER FIBER, BIDIRECTIONAL

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24621.

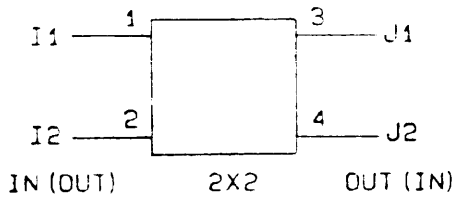


NOTES:
1. All dimensions are in mm.
2. Unless otherwise specified, tolerance is ±0.25 mm (.010).

FIGURE 1. Fiber optic coupler.

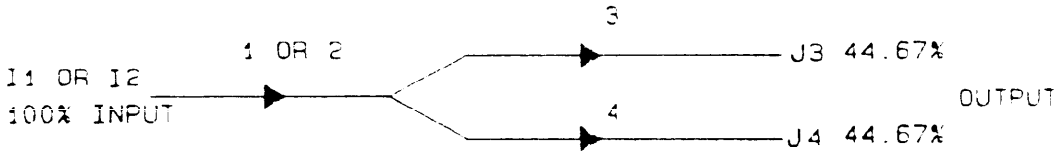
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

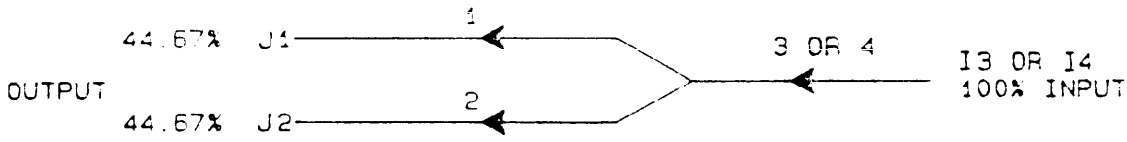
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$$P_{IJ} = \begin{matrix} P_{11} & P_{12} & P_{13} & P_{14} \\ P_{21} & P_{22} & P_{23} & P_{24} \\ P_{31} & P_{32} & P_{33} & P_{34} \\ P_{41} & P_{42} & P_{43} & P_{44} \end{matrix}$$

$$P_{IJ} = \begin{matrix} -30 & -30 & -3.5 & -3.5 \\ -30 & -30 & -3.5 & -3.5 \\ -3.5 & -3.5 & -30 & -30 \\ -3.5 & -3.5 & -30 & -30 \end{matrix}$$





NOTES:
1. I1 and J1 are illustrated as the dominant ports of the coupler.
2. Each matrix element specified here has a maximum allowed tolerance of ±0.25 dB.
3. Matrix elements above do not include the connector losses of the mating connector plug.

FIGURE 2. Transmission matrix derivation configuration.

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REQUIREMENTS:

Design and construction: Design and construction shall be in accordance with figure 1.

Cable pigtails: Cable pigtails shall be in accordance with MIL-C-85045/14, using 62.5/125 micrometer fiber surrounded by aramid strength members and an outer jacket.

Weight: Weight shall be less than or equal to 50 grams.

Color: The color shall be gray.

Temperature range: The operating temperature range shall be -46°C to +85°C. The storage temperature range shall be -62°C to +85°C.

Polarization sensitivity: Not applicable.

Fluid immersion: Not applicable.

Fiber dynamic strength: Not applicable.

Part or Identifying Number (PIN): The PIN shall be M24621/5-01.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Navy - AS, EC
Air Force - 11, 17, 19, 80, 99
DLA - ES

User activities:
Navy - CG, MC

Preparing activity:
Navy - SH

Agent:
DLA-ES

(Project 6005-0001-1)

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